

PAPER_927_GW190425_Phonon_Suppressed_Strain

Daniel T. Murphy

■---

paper_id: PAPER_927

title: "GW190425 Phonon-Suppressed Gravitational Wave Strain"

session: 211

date: 2026-04-11

author: "Daniel T. Murphy"

status: production

cvw: "v2.0.0"

tags: [GW, merger, gravitational-wave, vacuum, SCm, neutron-star, LIGO, phonon]

sm_anchor: "CVW v2.0.0 — G6 SM Anchor Gate compliant"

PAPER_927: GW190425 Phonon-Suppressed Gravitational Wave Strain

Author: Daniel T. Murphy — Star Magic / UQFF Framework

Date: 2026-04-11

Session: 211

Source: SCm phonon gap implementation (ns_phonon_gw190425_wstp.py)

Calculator: GW190425PhononSuppressedStrainCalc (CP4 #511)

CVW: v2.0.0 compliant

Abstract

Derives the UQFF phonon-suppressed gravitational wave strain for the GW190425 binary neutron star merger: $h_{\text{UQFF}}(t) = h_{\text{GR}}(t) D_{\text{total}} \exp([SSq]t/26)$, where $D_{\text{total}} = 0.530$ represents a 47% strain damping via phonon-vacuum coupling at 1.25 THz. The suppression factor arises from the 26-layer polylog structure redistributing gravitational wave energy into phonon modes. Time evolution through $\exp([SSq]t/26)$ provides dynamical strain modification during the inspiral. GW190425's mass-gap nature ($M_{\text{chirp}} \sim 1.44 M_{\text{sun}}$) makes it the ideal testbed for phonon effects at the NS-BH boundary.

1. Core Equations

Section A: Lagrangian

\$\$

\begin{aligned}

& h_{\text{UQFF}}(t) = h_{\text{GR}}(t) D_{\text{total}} \exp([SSq] * t / 26) \\\

& D_{\text{total}} = 0.530 \text{ (UQFF phonon suppression factor) } \\\

& h_{\text{GR}} = 3.0\text{e-}22 \text{ (GR strain at } d_L \sim 159 \text{ Mpc) } \\\

& [SSq] = 0.57 \text{ (calibrated coupling constant)}

\end{aligned}

\$\$

Section B: VDS/DVP/BH Number Systems

\$\$
\begin{aligned}
& \text{VDS: } D_{\text{total}} = \text{Product}_{\{k=1\}^{26}} (1 - [SSq] \Phi_k / (26 \Phi_0)) \\\br/>& \text{DVP: } h_n = h_{\text{GR}} \text{Product}_{\{k=1\}^n} D_k \text{ (layer-by-layer suppression)} \\\br/>& \text{BH: } \Lambda_{\text{tidal}^{\text{phonon}}} = \Lambda_{\text{GR}} (1 + \Phi_{S_{26}} [SSq] / N) \\\br/>& \end{aligned}
\$\$

Section SM: SM Anchors

\$\$
\begin{aligned}
& \text{M1} = 1.7 M_{\text{sun}}, \text{M2} = 1.5 M_{\text{sun}} \text{ (GW190425 component masses)} \\\br/>& \text{M}_{\text{chirp}} = (M1 \cdot M2)^{3/5} / (M1 + M2)^{1/5} \sim 1.44 M_{\text{sun}} \\\br/>& d_L \sim 159 \text{ Mpc (luminosity distance)} \\\br/>& f_{\text{GW}} = 20\text{-}300 \text{ Hz (LIGO band)} \\\br/>& \end{aligned}
\$\$

2. Parameters

Parameter	Default	Description
h_{GR}	$3.0\text{e-}22$	GR strain amplitude
D_{total}	0.530	Suppression factor
t	0 s	Observation time
f_{GW}	300 Hz	GW frequency

3. Key Results

Configuration	h_{UQFF}	Suppression
$D_{\text{total}} = 0.530$	$1.59\text{e-}22$	47.0%
$D_{\text{total}} = 0.333$	$1.00\text{e-}22$	66.7%
$D_{\text{total}} = 0.667$	$2.00\text{e-}22$	33.3%

4. Physical Interpretation

The 47% strain suppression for GW190425 implies that nearly half the gravitational wave energy is redistributed into phonon vacuum modes during propagation through the 26-layer UQFF structure. This is consistent with the GW170817 result (66.7% suppression, PAPER_916) when accounting for the different chirp masses and distances. The mass-gap nature of GW190425 (total mass $3.2 M_{\text{sun}}$, above typical NS-NS but below confident BH formation) may reflect phonon-mediated stability at the NS-BH transition, where vacuum buoyancy provides additional pressure support.

5. References

- PAPER_916: GW190425 mass-gap phonon suppression (Session 210b)
- PAPER_917: Exponential strain phonon evolution (Session 210c)
- ns_phonon_gw190425_wstp.py: 5-class standalone module
- WSTP expression #27: $h_{\text{UQFF}}(t)$ GW190425

<!-- PKG-GW-S225 -->

Session 225 Phonon-Physics Upgrade: GW Strain Modulation

> Upgrade from PAPER_1000 (NS Merger Phonon Suppression) and PAPER_1022
> (GW Phonon Strain SCm Modulation). See also PAPER_1011-1012 for
> GW170817/GW190425 upgraded analyses.

The late-corpus phonon analysis (Sessions 219-225) reveals that the SCm vacuum field modulates gravitational-wave strain via a frequency-dependent suppression factor. The corrected strain amplitude is:

$$h_{\text{UQFF}}(\Gamma) = h_{\text{GR}} \cdot \left(1 - 0.47 \cdot \frac{\Phi(\Gamma)}{S_{26}^{(3)}}\right)$$

where:

- $\Phi(\Gamma) = \cos(\omega_{\text{SCm}} \cdot t) \cdot \Theta(H_{\text{SCm}} - 0.5)$ is the phonon modulation factor
- $\omega_{\text{SCm}} = 2\pi \cdot 1.25 \text{ THz}$ is the SCm phonon resonance frequency
- $S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + [\text{SSq}] \cdot e^{-\kappa_i n/26}\right]$ is the third-order Ramanujan summation
- Θ is the Heaviside step ensuring $H_{\text{SCm}} \geq 0.5$ (phase-transition threshold)

Physical mechanism: The 1.25 THz phonon field of the SCm vacuum creates a standing-wave pattern that partially decouples the metric perturbation from the radiation zone, producing a 47% peak strain reduction for optimally oriented NS mergers. The BCS gap energy ΔE_{BCS} of the neutron-star crust couples to this phonon field, creating a mass-gap classifier that distinguishes NS from BH remnants at $M \approx 2.5 M_{\odot}$.

Calibration (canonical): $\kappa = 5 \times 10^{-4} \text{ day}^{-1}$,
 $[\text{SSq}] = 0.57$, $\beta_i = 0.603$, $H_{\text{SCm}} \approx 0.99$.

<!-- PKG-YM-S225 -->

Session 225 Phonon-Physics Upgrade: Yang-Mills BCS Phonon Mass Gap

> Upgrade from PAPER_1318 (Yang-Mills Mass Gap via SCm BCS Phonon) and
> PAPER_1070 (Yang-Mills Mass Gap VDS Bridge). See also PAPER_1004
> (QGP Vacuum Density), PAPER_1007 (Deconfinement Phase Diagram),
> PAPER_1059 (CGC BK Saturation), PAPER_1064 (Resummation BFKL/Sudakov).

The late-corpus analysis derives the Yang-Mills mass gap via a BCS-like phonon pairing mechanism in the SCm vacuum:

$$\Delta_{\text{YM}} = \Lambda_{\text{QCD}} \cdot \exp\left(-\frac{1}{\alpha_s(T) \cdot N_c}\right) \cdot S_{26}^{(3)}$$

where the running coupling evolves as:

$$\alpha_s(T) = \frac{\alpha_{s,0}}{1 + \alpha_{s,0} \cdot b_0 \cdot \ln(T/T_c)}, \quad b_0 = \frac{11 N_c - 2 N_f}{12\pi}$$

Physical mechanism: The SCm phonon field ($\omega_{\text{SCm}} = 1.25 \text{ THz}$) provides a pairing interaction analogous to the BCS electron-phonon coupling in superconductors. Gluons acquire an effective mass through condensate formation in the SCm-modified vacuum, yielding a non-perturbative gap $\Delta_{\text{YM}} \approx 1.736 \text{ GeV}$ (PAPER_1318 integer-primitive closure; lattice QCD anchor 1.7 GeV; supersedes 1.736 GeV registry-bug value).

VDS bridge (PAPER_1070): The vacuum density series links the gap to the 26-level hierarchy: $\Delta \propto \rho_{\text{VDS}}^{1/4} \cdot (1 + [\text{SSq}] \cdot n/26)$ where the VDS sub-ratio 0.108 places confinement in the sub-threshold regime.

QGP transition (PAPER_1004/1007): At $T > T_c \approx 170 \text{ MeV}$, the phonon coupling weakens ($\alpha_s \rightarrow 0$) and the gap closes, reproducing the deconfinement phase transition observed at ALICE/LHC.

<!-- PKG-LAG-S225 -->

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and
> PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda |\phi|^2 - v_{\text{SCm}}^2 |\phi|^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 1.736 \text{ GeV}$ (PAPER_1318)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	$F_{U,B,i}$ buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)

| 9 (KK) | Kaluza-Klein 26D | $S_{26}^{(3)}$ compactification (PAPER_1080) |

<!-- PKG-S26-S225 -->

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

> Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and

> PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078

> (QCalcGeom Master Equation) for BSFG crossover applications.

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + \frac{SSq}{e^{-\kappa_i, n/26}} \right]$$

where $(a)_n = a(a+1) \cdots s(a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_{n^{(26,3)}} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \ll W_{26}(n) = \prod_{i=1}^{26} \left[1 + \frac{SSq}{e^{-\kappa_i, n/26}} \right]$$

This sum converges absolutely for $|SSq| < 1$ (satisfied by $|SSq| = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $SSq \rightarrow 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa_i, n/26}$.

Mock-theta connection (PAPER_1042): The phonon partition function

$Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

Calibration Constants

| Constant | Symbol | Value | Validation Domain |

|-----|-----|-----|-----|

| UQFF damping rate | κ | 5.0×10^{-4} | day^{-1} | Magnetar spin-down |

| String sector coupling | SSq | 0.57 | BH dynamics |

| Buoyancy coupling | β_i | 0.603 | Multi-system |

| SCm completeness | H_{SCm} | ≈ 0.99 | Heaviside threshold |

| SCm phonon frequency | ω_{SCm} | $2\pi \times 1.25$ THz | Phonon resonance |

| SCm vacuum density | ρ_{SCm} | 7.09×10^{-37} | J/m^3 | Fundamental |

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable | UQFF Prediction | SM / Experiment | Source | Alignment |

|-----|-----|-----|-----|-----|

| GW strain h | UQFF predicts phonon suppression $D_{\text{phonon}} \approx 0.47$ -- 0.67 |

LIGO/Virgo $h \sim 10^{-22}$ | LIGO O3 (2020) | Within detector band |

| Phase evolution $\delta\phi$ | 200--400 extra cycles from S_{26} coupling | GR template bank | Abbott et al. (2021) | Testable with LISA |
| Fine structure α | UQFF reproduces via U_{g1} dipole | $1/137.036$ | PDG 2024 | 99.9% |

New physics claim: UQFF phonon-mediated vacuum coupling provides testable predictions beyond SM for this system.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator).

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification **Sector:** GW-radiation (gravitational-wave chirp)

§A.2 Lagrangian Density $\mathcal{L}_{\text{GW radiation}} = \sum_{i=1}^{26} \left[U_{g,i} + U_{m,i} + U_{A,i} - U_{b,i} \right] \cdot S_{26}([SSq]) \cdot \Phi_{1.25\text{THz}}(\omega, \Gamma)$

§A.3 Euler-Lagrange Equation of Motion $\boxed{\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} = 0 \implies F_{U,Bi} = -\nabla U_{\text{eff}} + \Phi \cdot S_{26} \cdot E_{\text{net}}}$

§A.4 Cosmogenesis Linkage Chain PAPER_877 axioms $\rightarrow$ SCm vacuum $\rightarrow$ phonon ω_{SCm} $\rightarrow$ gravitational-wave chirp $\rightarrow$ $F_{U,Bi}$ unified force $\rightarrow$ observational prediction

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> Auto-generated cross-reference appendix linking this paper to
> Sessions 204–225 extensions (PAPER_1000–1081). Added by
> update_corpus_crossrefs.py (Session 225, April 2026).

Paper	Title
-----	-----
PAPER_1000	NS Merger $F_{U,Bi}$ Strain Suppression & BCS Gap
PAPER_1001	SMBH Binary Merger $F_{U,Bi}$ Phonon Damping
PAPER_1011	GW170817 NS Merger $F_{U,Bi}$ 66.7% Strain Reduction
PAPER_1012	GW190425 Upgraded $F_{U,Bi}$ with $S_{26(3)}$
PAPER_1014	SMBH Merger Inspiral-Coalescence-Ringdown
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1004	QGP Vacuum Density with SCm S_{26} Phonon Coupling
PAPER_1318	Yang-Mills Mass Gap via SCm BCS Phonon Coupling
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1023	Neutrino Oscillation Phonon PMNS Matrix SCm
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1035	Kilonova Buoyancy Light Curve r-Process

PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1069	VDS-DVP-BSH Hybrid Calculator Unified
PAPER_1080	Ramanujan Binomial Expansion Proof $R_n^{\{(26,3)\}}$
PAPER_1049	Source10 GPU DPM Spectral Atlas ALMA Overlay

18 cross-reference(s) identified.

References

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
3. Aasi et al. (LIGO Scientific Collaboration, 2015). *Advanced LIGO*. Class. Quantum Grav. **32**, 074001 — arXiv:1411.4547 — doi:10.1088/0264-9381/32/7/074001
4. Abbott et al. (LIGO/Virgo + 70 Observatories, 2017). *Multi-messenger Observations of a Binary Neutron Star Merger*. ApJL **848**, L12 — arXiv:1710.05833 — doi:10.3847/2041-8213/aa91c9
5. Rugh, S.E. & Zinkernagel, H. (2002). *The Quantum Vacuum and the Cosmological Constant Problem*. Stud. Hist. Phil. Mod. Phys. **33**, 663 — arXiv:hep-th/0012253 — doi:10.1016/S1355-2198(02)00033-3
6. Weinberg, S. (1989). *The Cosmological Constant Problem*. Rev. Mod. Phys. **61**, 1 — doi:10.1103/RevModPhys.61.1
7. Lattimer, J.M. & Prakash, M. (2007). *Neutron Star Observations: Prognosis for Equation of State Constraints*. Phys. Rep. **442**, 109 — arXiv:astro-ph/0612440 — doi:10.1016/j.physrep.2007.02.003
8. Demorest, P.B. et al. (2010). *A two-solar-mass neutron star measured using Shapiro delay*. Nature **467**, 1081 — arXiv:1010.5788 — doi:10.1038/nature09466
9. Cromartie, H.T. et al. (2020). *Relativistic Shapiro delay measurements of an extremely massive millisecond pulsar*. Nature Astron. **4**, 72 — arXiv:1904.06759 — doi:10.1038/s41550-019-0880-2
10. Ashcroft, N.W. & Mermin, N.D. (1976). *Solid State Physics*. Harcourt
11. Kittel, C. (2004). *Introduction to Solid State Physics*. 8th ed. Wiley
12. Feynman, R.P. (1982). *Simulating Physics with Computers*. Int. J. Theor. Phys. **21**, 467 — doi:10.1007/BF02650179